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An integrated magneto-electrochemical device for the rapid profiling of tumour extracellular vesicles from blood plasma

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Assays for cancer diagnosis via the analysis of biomarkers on circulating extracellular vesicles (EVs) typically have lengthy sample workups, limited throughput or insufficient sensitivity, or do not use clinically validated biomarkers. Here we report the development and performance of a 96-well assay that integrates the enrichment of EVs by antibody-coated magnetic beads and the electrochemical detection, in less than one hour of total assay time, of EV-bound proteins after enzymatic amplification. By using the assay with a combination of antibodies for clinically relevant tumour biomarkers (EGFR, EpCAM, CD24 and GPA33) of colorectal cancer (CRC), we classified plasma samples from 102 patients with CRC and 40 non-CRC controls with accuracies of more than 96%, prospectively assessed a cohort of 90 patients, for whom the burden of tumour EVs was predictive of five-year disease-free survival, and longitudinally analysed plasma from 11 patients, for whom the EV burden declined after surgery and increased on relapse. Rapid assays for the detection of combinations of tumour biomarkers in plasma EVs may aid cancer detection and patient monitoring.

The assessment of circulating biomarkers, also known as liquid biopsy, is an emerging approach to obtaining molecular information about patients' tumours through repeated yet minimally invasive sampling[1–4]. One appealing target for such assessment is extracellular vesicles (EVs)—that is, membrane particles secreted by cells[5,6]. In addition to being generally abundant and stable, EVs have been reported to carry biomolecules (such as proteins[7,8], nucleic acids[9–11] and lipids[12]) of parent cells. The analysis of tumour-derived EVs in particular has considerable potential to reveal the dynamic status of tumours[13–15] and thereby improve current cancer diagnostics[10,11,14–20]. Establishing clinical EV tests, however, faces several technical challenges, namely (1) laborious manual sample preparation; (2) limited sensitivity and throughput of existing tools; and (3) high cost of test equipment or assays (such as sequencing). In short, for EV diagnostics to become clinically useful, new integrative methods for the isolation and molecular analyses of EVs are needed, ideally ones that are amenable to high-throughput operations[21]. It is equally important to analyse large clinical samples to establish robust EV biomarker baselines for disease status.

Here we report the development of a clinically adoptable high-throughput EV analysis technology termed high-throughput integrated magneto-electrochemical extracellular vesicle (HiMEX). This technology streamlines EV analyses by combining EV enrichment and electrochemical detection in a single assay, which enables EV protein profiling directly from clinical samples; the total assay time was less than 1 hour (in a 96-well format) and the analytical signal was read out in parallel from all (96) detection probes. We then applied HiMEX to the analysis of clinical samples, and show the practical advantages of HiMEX. Specifically, we analysed blood samples from 102 patients with colorectal cancer (CRC) as they underwent surgery and chemotherapy, plus 40 non-CRC controls (that is, healthy volunteers or patients with non-CRC diseases such as appendicitis or small bowel internal herniation) ($n = 142$ in total). HiMEX analyses revealed the diverse potentials of EVs for CRC management. (1) EVs that reflected the key protein signatures of parental tumours were present in the circulation, and the detection of such CRC-derived EVs (CRC-EVs) led to highly accurate cancer diagnoses (overall accuracy >96%). (2) Serial changes in CRC-EVs could be related to the patients' treatment responses. Notably, levels of CRC-EV decreased in all patients after curative surgery but rebounded from each patient's baseline values with tumour recurrence. (3) Preoperative CRC-EV levels showed significant correlation with patients' five-year disease-free survival (DFS) ($n = 90$), which effectively categorizes patients into high- and low-risk groups. These outcomes have implications for timely, better-informed

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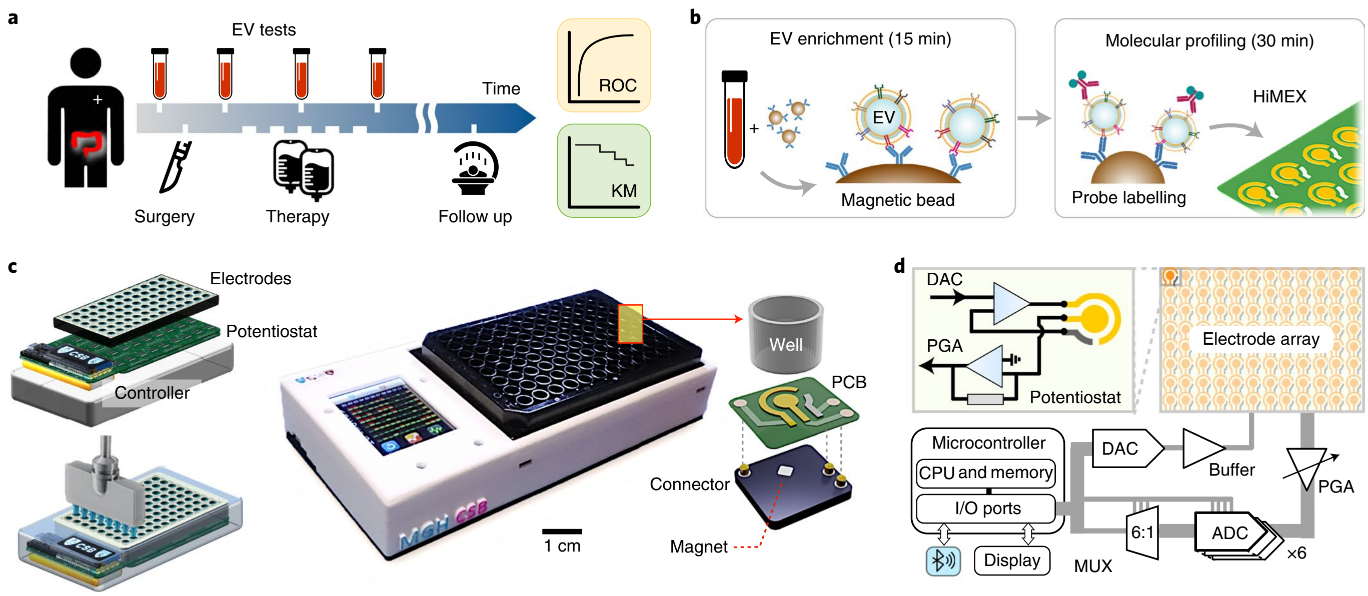

Fig. 1 | HiMEX approach for clinical EV analyses. a, Study design. We collected blood samples from patients with CRC ($n=102$) during their standard clinical care. EVs and conventional markers (such as CEA and CA19-9) were analysed from blood samples and cross-compared with clinical outcomes, including immunohistology, radiological reports and survival. KM, Kaplan–Meier; ROC, receiver operating characteristic. **b**, Two-step HiMEX assay protocol. EVs are enriched via immunomagnetic capture. Bead-bound EVs are then labelled with probing antibodies for signal generation by electrochemical reaction. The assay is simple and fast (<1 h), and analyses plasma samples directly without requiring purification steps. **c**, We developed a compact HiMEX reader with a touchscreen interface. The reader accommodated an array of 96 electrodes in a conventional 96-well plate. Under each electrode were push-pin connectors to make electrical contacts and a magnet to concentrate bead-bound EVs. PCB, printed circuit board. **d**, The HiMEX reader was designed to carry out 96 parallel measurements within 2 min. Each electrode was connected to its own potentiostat. A microcontroller applied electrical potential through a digital-to-analogue converter (DAC), initiating electrochemical reaction. Resulting currents from the electrode were read out by an analogue-to-digital converter (ADC) via rapid multiplexing (MUX). The microcontroller processed and displayed data on a touchscreen. The reader also communicated with an external device (for example, a smartphone) for data logging. PGA, programmable gain amplifier; I/O, input and output.

cancer care, and broaden the clinical utility of EVs for CRC diagnosis, recurrence monitoring and prognosis.

Results

HiMEX approach for clinical EV tests. Our study aimed to evaluate HiMEX for applications in CRC diagnosis, treatment monitoring and prognosis (Fig. 1a and Supplementary Fig. 1). Overall, we analysed plasma sample from 102 patients with CRC and 40 non-CRC controls ($n=142$ total). For CRC diagnostics, we first defined a CRC-EV signature based on published results, in vitro studies and tissue immunohistochemistry. We next measured these markers in plasma EVs. A total of 131 patient samples were used: a training cohort consisting of 25 non-CRC controls and 58 patients with CRC; and a testing cohort of 15 non-CRC controls and 33 patients with CRC. For patient monitoring, we followed an additional 11 patients with CRC as they underwent standard clinical care. Serial blood samples were collected at defined time points (that is, before and after surgery and after chemotherapy) and analysed for CRC-EV markers. The EV profiling results were then compared with clinical information, including the levels of conventional tumour markers (such as carcinoembryonic antigen (CEA) and carbohydrate antigen (CA19-9))[22,23], radiological assessment and treatment outcomes. Finally, for the prognostic use of EV profiling, we correlated five-year DFS with pre-surgery CRC-EV levels among 90 patients with CRC.

To streamline EV analyses, we optimized the HiMEX technology, integrating sample preparation and measurement within a single assay (Fig. 1b). We used immunomagnetic separation to enrich target-specific EVs, which enabled the rapid (~15 min) isolation of EVs directly from native samples without additional manipulation. We then harnessed magnetic beads as substrates for subsequent signal generation; bead-bound EVs were labelled with probing antibodies functionalized with catalysing enzymes for electrochemical reactions[16,24]. The HiMEX strategy had the following merits: (1) sample handling was simple (that is, magnetic pulling), with no extensive purification of EVs required; (2) the assay benefitted from fast binding kinetics and high efficiency in EV capture and labelling (around 30 min in total); (3) the analytical signal was amplified by enzymatic reaction; and (4) the detection modality (that is, electrical measurements) was amenable to scale-up for high-throughput measurements.

HiMEX detection system for parallel measurements. To exploit the unique technical advantages of HiMEX, we developed a compact system capable of parallel measurements (Fig. 1c). The system was compatible with a 96-well plate to enable batch processing. EV capture and labelling were performed in a conventional 96-well plate; a custom-designed magnet array (Supplementary Fig. 2a) was used for bead handling (for example, washing and buffer exchange). EV-bound beads were then spotted on a 12×8 probe array (Supplementary Fig. 2b) that was loaded to the detection device. Each probe in the array, which contained three electrodes (that is, working, counter and reference), was connected to a potentiostat by quick push-pin connectors and was placed over a small embedded magnet to concentrate magnetic beads (Fig. 1c and Supplementary Fig. 2c). The detection system also featured a touchscreen interface for device control and data display, and wirelessly communicated (via Bluetooth) with external servers for data storage.

As an analytical signal, electrical currents between working and counter electrodes were monitored while a constant voltage

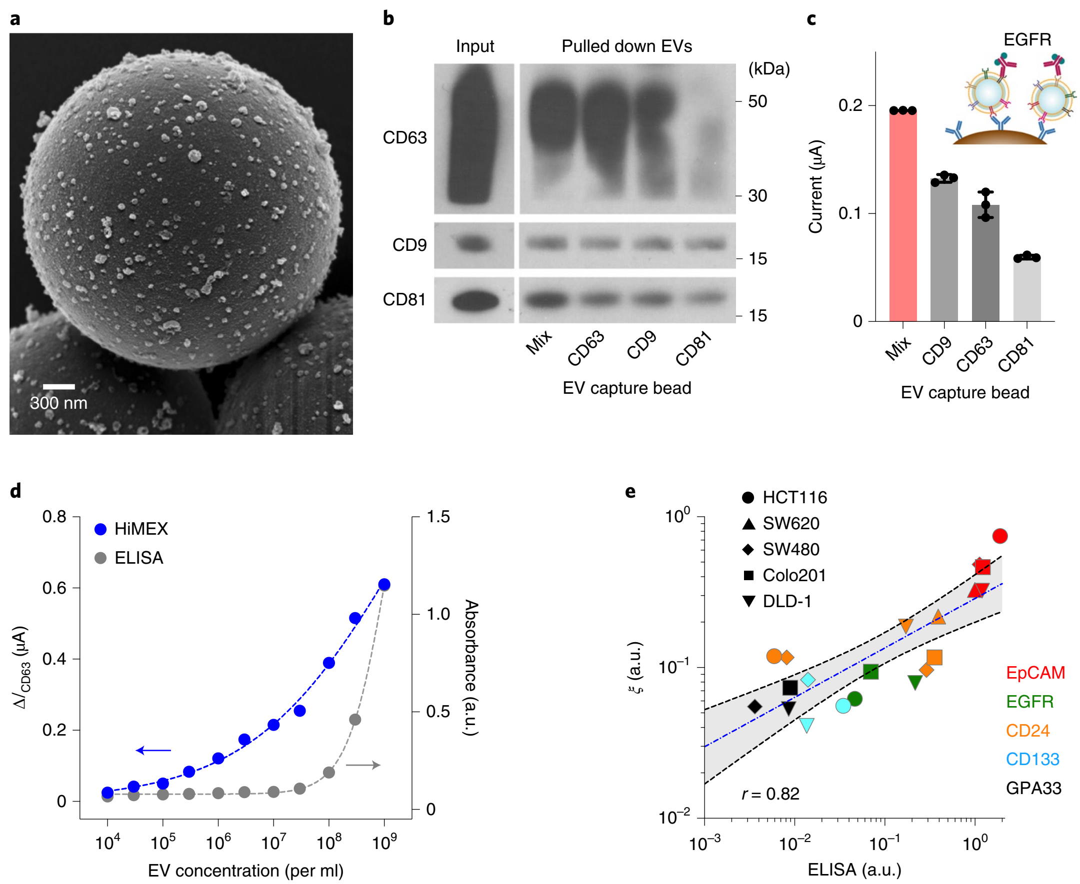

Fig. 2 | HiMEX assay optimization and characterization. a, Scanning electron micrograph of microbeads after incubation with EV samples. The beads (diameter, 3 μm), functionalized with antibodies against CD63, captured EVs isolated from a cell culture medium (HCT116 cell line). A representative image is selected from technical duplicate samples. **b**, Three types of magnetic bead, with each type specific to a different tetraspanin, and their mixture (Mix), were used to capture EVs. Key tetraspanin expression was then measured on captured EVs. The bead cocktail most efficiently overcame EV heterogeneity. A representative image is selected from duplicate measurements. Full western blot images are shown in the Supplementary Information. **c**, EVs (10^7 ml^{-1}) from HT29 cell lines were captured and further labelled with EGFR antibody. Mixed bead types led to the highest HiMEX signal. Data are mean ± s.d. from technical triplicates. **d**, The HiMEX assay had superior sensitivity and a wider dynamic range than conventional enzyme-linked immunosorbent assay (ELISA). The LOD was approximately 10^4 EVs ml^{-1} for HiMEX and 10^7 EVs ml^{-1} for ELISA. a.u., arbitrary units. EV numbers were estimated from nanoparticle tracking analysis. The data points represent mean values from technical duplicates. **e**, EVs from different CRC cell lines were profiled to a set of protein markers. The HiMEX results showed a good match (Pearson correlation coefficient, $r = 0.82$) with those from ELISA. To obtain the HiMEX expression (ξ) of a target protein marker (M), the marker-associated current level was normalized against the loading control, the current level of CD63. For ELISA, the same amount of EVs (10^9 EVs ml^{-1}) was used for each marker. Data were plotted on log scales to display small values better. The blue-dashed line indicates the best linear fit in the log–log scale, and the shaded grey area denotes the 95% confidence band. HiMEX data represent mean values from technical duplicates.

was applied between working and reference electrodes. To enable fast readouts, we designed the system electronics to rapidly (100 Hz) poll each probe during measurements (Fig. 1d): an embedded microcontroller sequentially accessed individual probes by six analogue-to-digital converters and a 6-to-1 multiplexer. This scheme effectively executed real-time, parallel measurements (Supplementary Fig. 2d). The total readout time for all 96 probes was less than 2 min.

We benchmarked the system performance using samples with varying concentrations of $K_4Fe(CN)_6$. We particularly focused on assessing the reproducibility among 96 probes. For a given $K_4Fe(CN)_6$ concentration, different probes in the array reported a consistent signal (Supplementary Fig. 3a). Four standard curves, generated by parallel measurement of $K_4Fe(CN)_6$ dilution series, showed an excellent linear relationship between current levels and $K_4Fe(CN)_6$ concentrations (Supplementary Fig. 3b). Notably, these curves were statistically identical, which demonstrates high fidelity among probes (that is, low intra-probe variations) for the parallel detection.

HiMEX assay characterization. We next optimized key assay steps, namely EV capture and electrochemical detection, to maximize the analytical signal. For EV capture, we prepared three types of magnetic bead, each specific to one of the tetraspanins (that is, CD63, CD9 and CD81) enriched in EVs. When mixed with EVs, these beads were effectively covered with vesicles (Fig. 2a); control beads conjugated with IgG antibodies showed minimal non-specific binding (Supplementary Fig. 4). We further used either single types or a cocktail of capture beads for EV pull-down and compared levels of EV tetraspanin (Fig. 2b). A mixture of capture beads (CD63, CD9 and CD81) resulted in a higher signal across all tetraspanin levels than any single-marker bead alone (Fig. 2b and Supplementary Fig. 5a), which presumably reflects EV heterogeneity[25–27]. Single EV imaging (Supplementary Fig. 6) supported the observation; the number of EVs visible by microscopy was the highest when the cocktail of CD63, CD9 and CD81 antibodies was used for fluorescent staining. The immunomagnetic capture was also efficient in plasma (Supplementary Fig. 5b); on the basis of CD63 levels before and after immunocapture, the pull-down yield was estimated to be 82%.

To generate the detection signal, we used a probing antibody to label captured EVs with an oxidizing enzyme (horseradish peroxidase (HRP)) and mixed the conjugates with chromogenic electron mediators (3,3′,5,5′-tetramethylbenzidine (TMB)). The redox reaction quickly reached an equilibrium, with electrical currents plateauing within 1 min (Supplementary Fig. 7). We recorded the current between 50 and 55 s after initiating the reaction and used the average value (I) as an analytical metric. Because it consistently produced the maximum HiMEX signal (Fig. 2c), we used the three-bead cocktail for EV capture throughout the rest of the study.

Our next step was to characterize the analytical capacities of HiMEX. First, we prepared samples with varying concentrations of EV. Using the CD63, CD9 and CD81 bead mixture, we captured EVs and labelled them with CD63; we generated control samples by labelling captured EVs with IgG antibodies. These samples were then measured using the HiMEX detector (Supplementary Fig. 7). We used the net current $\Delta I_{CD63} = I_{CD63} - I_{IgG}$ as an analytical metric to minimize the effect of common-mode errors (for example, temperature fluctuation and non-specific bindings). From the titration results (Fig. 2d), the limit of detection (LOD) of the HiMEX technology was estimated to be around 10^4 EVs, more than 1,000-fold lower than by conventional ELISA assay. HiMEX also had a much wider dynamic range (approximately 10^5 EVs) than ELISA assay (fewer than 10^3 EVs). We next assessed assay reproducibility. On three different days, we prepared EV samples and probed them for CD63, CD9 and CD81 using the HiMEX array. For a given marker, the observed signals were statistically identical on different days (Supplementary Fig. 8). We further compared signal levels with EVs spiked in either plasma or serum (Supplementary Fig. 9) and observed similar signal changes in both sample types. We opted for plasma as it is considered the physiological medium for EVs, and contains fewer platelet-derived vesicles[28].

To test whether EVs captured by the bead mixture contained CRC-relevant markers, we performed a competitive assay (Supplementary Fig. 10a). Samples were prepared by spiking EVs from a CRC cell line (SW480 cells; positive for EpCAM and CD24) into human plasma. EV immunomagnetic capture was performed in the presence of excess, free-floating capture antibodies (a mixture of CD63, CD9 and CD81). We then probed bead-bound EVs for the expression of CD63, EpCAM and CD24. The measured HiMEX signal decreased as the concentration of free antibodies increased (Supplementary Fig. 10b), which indicates the presence of CRC markers on the bead-captured EVs.

For a quantitative HiMEX assay, we established the following protocol. For a given marker (M) of interest, we measured the HiMEX signal ($\Delta I_M = I_M - I_{IgG}$) after EV capture and labelling. As an EV-loading control, we also measured ΔI_{CD63} using an aliquot from the same sample. The expression level (ξ_M) of the target marker was then estimated by scaling I_M against the loading control ($\xi_M = \Delta I_M / \Delta I_{CD63}$). We applied the method to profile EVs for different protein markers. The results showed a good linear correlation (Pearson $r = 0.82$) with ELISA (Fig. 2e).

High-throughput screening of cell-line derived EVs. To choose the initial EV markers relevant for CRC detection, we applied a custom-designed bioinformatic pipeline to public databases (Methods and Supplementary Fig. 11). From the Human Protein Atlas, we retrieved protein expression profiles for CRC and healthy tissues and chose markers that were overexpressed in CRC. This collection was then cross-referenced with UniProt annotations to select proteins that contained an extracellular domain. We further narrowed down the list by applying two criteria: (1) reported marker presence in EVs (Vesiclepedia database); and (2) availability of ELISA-compatible antibodies (Supplementary Table 1). This algorithm selected nine markers (CD44, B7-H3, EGFR, ABCG2, GPA33, EpCAM, HER2, MUC1 and MET). We augmented the list by including markers that were found to be overexpressed in CRC[29–34] (CD44v6, CD24, CD166, STEAP1 and ALDH1) and those used for the monitoring of drug resistance[35–37] (MRP1, MDR1, TS and CD133).

We then measured the candidate markers across EVs derived from CRC cells, using the high-throughput capacity of HiMEX. This preclinical study focused on (1) investigating the correlation in marker expression between EVs and their originating cells; and (2) checking the presence of candidate markers in EVs. The cell-line panel included cells that represent different CRC stages[38] (SW480 cells, Dukes' classification type B; DLD-1 and SW620 cells, type C; Colo201 cells, type D) as well as drug-resistant (irinotecan) phenotypes (HT29 and HCT116). As a negative control, we profiled EVs from a healthy colon cell line (CCD-18Co) as well as plasma from healthy donors; the inclusion of healthy plasma samples was justifiable as the analytical goal was to identify markers that were overexpressed in CRC-EVs. After collecting EVs, we profiled them by HiMEX (Supplementary Fig. 12a), and cellular expression was measured by flow cytometry. Overall, the expression profiles of the selected markers for EVs and their parent cells matched closely ($r = 0.89$) (Supplementary Fig. 12b), which supports the use of EVs as cellular surrogates[7,10,14,17,39]. For further testing with clinical EV samples, we chose EpCAM, EGFR, CD24 and CD133, on the basis of their relatively higher expression in CRC cell lines (Supplementary Fig. 12c). We also included GPA33 for its clinical relevance in CRC diagnostics[40].

Defining EV protein signature for CRC diagnostics in clinical samples. We next applied HiMEX to detect CRC EVs in clinical

Fig. 3 | EV profiling for CRC detection. a, Images of tumour tissue (left) and plasma EV analyses (right). Three key CRC protein markers (EpCAM, EGFR and CD24) were assessed in tumour tissues (immunohistochemistry) and blood samples (HiMEX) from each patient. Expression profiles showed a qualitative match. Data from two representative patients are shown. Data are mean ± s.e.m. from technical triplicate measurements. Full images of these samples are shown in the Supplementary Information. **b**, Tumour tissue (left) and plasma EV samples (right) from 12 patients with CRC were analysed for expression of EpCAM, EGFR and CD24. Tissue staining was graded as the fraction of marker-positive cells among total cancer cells. The pathological score and the EV expression profile showed significant correlation (Spearman rank coefficient $\rho_s = 0.65$, $P < 0.0001$, two-sided t-test). **c**, HiMEX analyses for CRC diagnosis. Top: as a training set, plasma samples from 58 patients with CRC before surgery and 25 healthy controls were analysed. The expression of five CRC markers (EpCAM, EGFR, CD24, CD133 and GPA33) was measured by HiMEX. Bottom: the diagnostic metric, EV_{CRC}, was determined as a weighted sum of four marker levels (EpCAM, EGFR, CD24 and GPA33) by logistic regression. **d**, The average level for each of the indicated markers was higher in patients with CRC than in healthy controls. The marker distribution, however, overlapped between patients and controls, which reduces the classification power of single markers. **e**, For each CRC marker and EV_{CRC}, ROC curves were constructed. The area under the curve (AUC) of EV_{CRC} was 0.98, which is significantly larger than those of single markers (all $P < 0.001$). Cut-off levels that maximized the sum of sensitivity and specificity were determined from the ROC curves. **f**,**g**, EV_{CRC} effectively differentiated patients with CRC from controls ($P = 0.0009$, unpaired two-sided t-test) (**f**). In the training cohort, the diagnostic accuracy was 98%. The cut-off value from the EV_{CRC} ROC curve was 4.96 (**g**).

samples. We first examined whether circulating EVs from patients with CRC contain key protein markers found in CRC tissue. From a pilot cohort of 12 patients with CRC, we obtained both preoperative blood samples and tissue specimens during surgery. We assessed the expression of the top three markers (EpCAM, EGFR and CD24) identified from the cell-line study. Tissue specimens were stained (immunohistochemistry) for these markers, and the expression was graded by a pathologist as a fraction of marker-positive cells

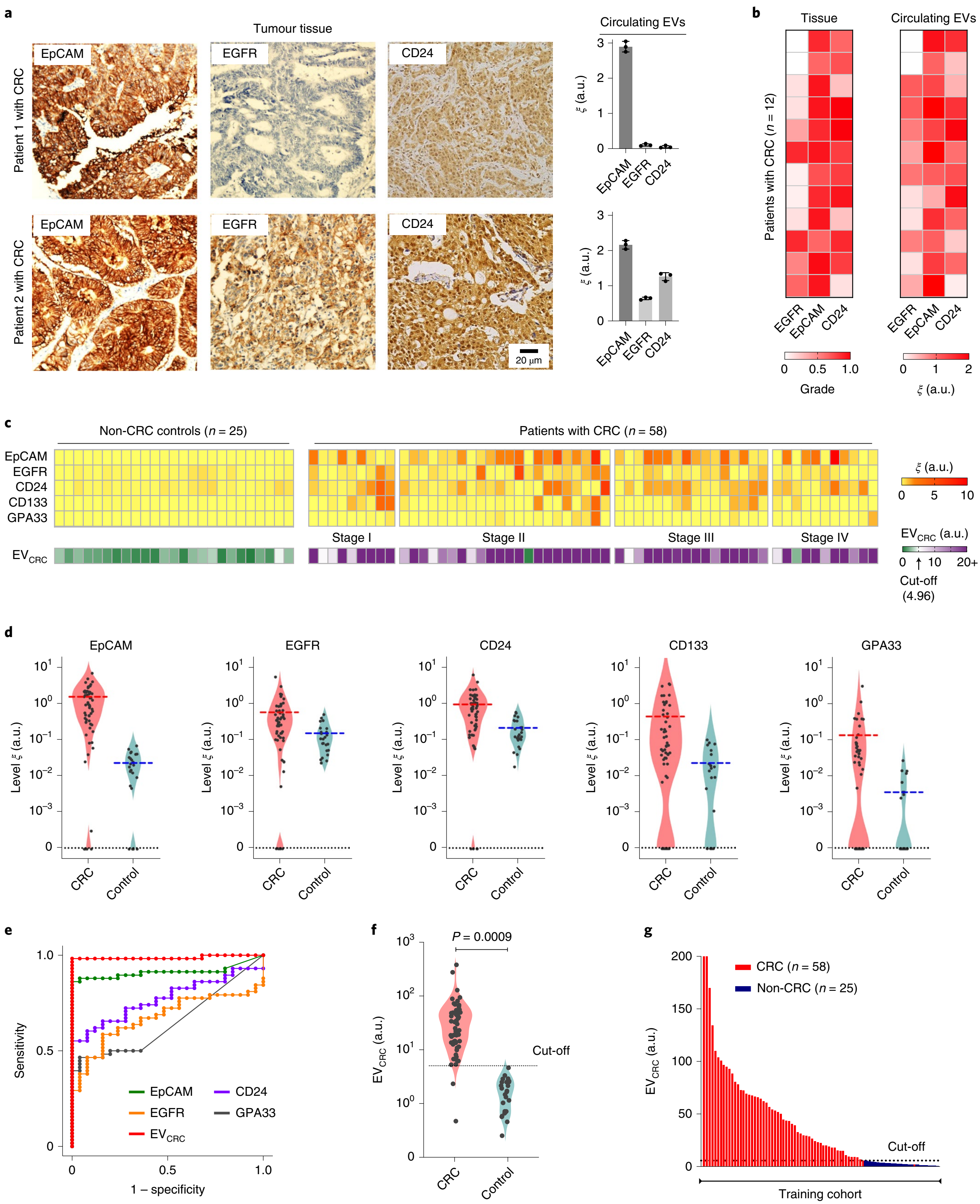

Table 1 | Clinical information of cohorts involved in CRC diagnostics

	Training cohort		Testing cohort		Total
Case	Non-CRC	CRC	Non-CRC	CRC	
	25	58	15	33	131
Age					
Median	40	66	59	63	61
Range	18–80	38–80	25–76	34–82	18–82
Sex					
Male	14 (56%)	35 (60%)	10 (67%)	19 (58%)	78 (60%)
Female	11 (44%)	23 (40%)	5 (33%)	14 (42%)	53 (40%)
Stage					
I	-	9 (16%)	-	1 (3%)	10 (11%)
II	-	22 (38%)	-	7 (21%)	29 (32%)
III	-	16 (28%)	-	21 (64%)	37 (41%)
IV	-	11 (18%)	-	4 (12%)	15 (16%)
Serum marker (median)[a]					
CA19-9 ($\mathrm{IU\,ml^{-1}}$)	6.9 (0.3–24.9)	17.8 (0.5–2,167)	4.2 (0.11–18.8)	14.7 (0.8–188.7)	
CEA ($\mathrm{ng\,ml^{-1}}$)	1.4 (0.4–5.2)	3.5 (0.4–7,671)	1.5 (0.2–4.8)	5.1 (0.4–37.8)	

[a]Ranges of marker concentrations are denoted in parentheses.

among total cancer cells (from 0 to 1 with the increment of 0.1) (Methods). EVs were screened for the same markers via HiMEX (Fig. 3a). Overall, marker expression between tissue and EV samples displayed good concordance (Fig. 3b and Supplementary Fig. 13) and a positive correlation (Spearman's rank coefficient $\rho_s = 0.65$, $P < 0.0001$). These results indicated the presence of CRC-derived EVs in the circulation, and thereby support the rationale of assessing tumour molecular status by EV profiling.

To establish CRC-EV diagnostics, we expanded the cohort and used the full marker set in EV profiling (Table 1). Blood was collected from patients with CRC before surgery. Control samples were obtained from healthy donors as well as patients with non-CRC diseases. All blood samples were collected and consistently processed as per our established protocol (Methods). Circulating EVs were assessed by HiMEX for levels of EpCAM, EGFR, CD133, GPA33, CD24 and CD63. Aliquots of samples were also assayed for the clinically relevant serum markers CEA and CA19-9.

Figure 3c summarizes EV marker expression profiles from the initial training cohort (58 patients with CRC, 25 non-CRC controls). The average levels of all five markers were higher in patients with CRC than in controls (Fig. 3d). However, the two distributions overlapped, which makes classification based on a single marker difficult. We therefore considered multi-marker combinations. Specifically, we used weighted sums of EV markers, in which optimal weights were determined by logistic regression (Methods). For both individual markers and combinations thereof, we constructed ROC curves (Fig. 3e and Supplementary Fig. 14). In each ROC curve, we determined a cut-off value that maximized the sum of sensitivity and specificity[41]. The combination of EGFR, EpCAM, CD24 and GPA33 was found to be optimal; this metric was thus defined as the EV_{CRC} score. The AUC of EV_{CRC} was significantly higher (all $P < 0.001$, DeLong's test) than those of the individual EV markers (Fig. 3e) and two- or three-marker combinations. Compared with the five-marker combination, EV_{CRC} showed statistically identical ($P = 0.17$, DeLong's test) classification performance. Overall, EV_{CRC} with the cut-off value of 4.96 showed a high diagnostic power (Fig. 3f,g) with the training cohort, achieving a detection sensitivity of 97%, specificity of 100%, and accuracy of 98% (Methods and Table 2).

We further tested the statistical model by analysing samples from a prospective testing cohort (33 patients with preoperative CRC and 15 non-CRC controls) (Fig. 4a). When the same cut-off values from the training sets were applied (Fig. 4b,c), EV_{CRC} maintained excellent diagnostic statistics (sensitivity, 94%; specificity, 100%; accuracy, 96%). The diagnostic accuracy of EV_{CRC} was superior to that of conventional serum markers (Supplementary Fig. 15a), presumably due to the CRC-specific nature of EV analyses. EV_{CRC} showed no significant correlation (Supplementary Fig. 15b) with CEA ($\rho_s = -0.12$, $P = 0.24$) or CA19-9 ($\rho_s = -0.11$, $P = 0.29$); combining either of these serum markers with EV_{CRC} (by logistical regression) did not significantly improve the AUC ($P = 0.20$ for the addition of CEA; $P = 0.18$ for the addition of CA19-9, DeLong's test).

Longitudinal CRC-EV monitoring during clinical care. Next, we tracked EV profile changes as patients underwent standard clinical care. We first compared EV levels before and after surgery (Fig. 5a). Blood samples ($n = 13$) were collected 24 h before surgery and within one week thereafter. Overall EV loads showed no significant changes ($P = 0.63$, paired t-test) or directionality. By contrast, EV_{CRC} decreased in all patients ($P < 0.0001$, paired t-test), presumably due to the presence of fewer CRC-derived EVs after curative tumour resection. The reasoning was further supported when we analysed blood samples of patients ($n = 4$) who underwent abdominal surgery for non-CRC diseases (for example, appendicitis and small bowel internal herniation). For these patients, EV_{CRC} values remained statistically identical ($P = 0.77$, paired t-test) before and after surgery (Supplementary Fig. 16). On the patient-wide average, levels of CEA and CA19-9 decreased after surgery, but in pairwise comparisons (pre- and post-surgery) the downward trend was not statistically significant ($P = 0.19$ for CEA and $P = 0.19$ for CA19-9, paired t-test).

To ascertain temporality better, we monitored additional patients with CRC ($n = 11$) for six months after surgery. Eight patients, either at stage II with risk factors (inadequate lymph node sampling, perforation or lymphovascular invasion) or at stage III, received adjuvant treatment (FOLFOX chemotherapy) with 5-fluorouracil (5-FU) plus leucovorin and oxaliplatin. The remaining three patients, all at stage II, were excluded from therapy (see Supplementary Table 2 for patient details). For treated patients, blood samples were collected before and after therapy; for non-treated patients, samples were collected around 6 months after surgery. The tumour recurrence status was confirmed later by either surgical resection or radiological detection of lesions that increased in size over time. Among the treated patients ($n = 8$), four were classified as non-recurrent and the rest as recurrent; tumours recurred in all non-treated patients ($n = 3$). Blinded to these classifications, we performed HiMEX EV profiling. In all patients, EV_{CRC} values decreased after surgery (Supplementary Fig. 17). From the post-surgery EV_{CRC} baseline, EV_{CRC} increased in all recurrent patients regardless of the treatment status. By contrast, EV_{CRC} decreased or became stable in non-recurrent patients (Fig. 5b). Serial changes in EV_{CRC} (ΔEV_{CRC}) thus showed potential as an indicator of short-term tumour relapse ($P = 0.0004$, unpaired two-sided t-test) (Supplementary Fig. 18). Changes in levels of CEA ($P = 0.13$, t-test) and CA19-9 ($P = 0.76$, t-test) were similar regardless of recurrence status.

EV capture and RNA analyses. The immunomagnetic capture used in HiMEX enabled us to enrich EVs not only for CRC protein detection (HiMEX) but also for other molecular assays. We applied this technique to aid in EV-RNA analyses in blood samples collected after adjuvant chemotherapy. As in the HiMEX assay, we used a cocktail of tetraspanin magnetic beads to capture EVs. Plasma

Table 2 | Summary of CRC diagnostic statistics for conventional and EV markers

Markers		Cut-off	Training cohort ($n=83$)				Testing cohort ($n=48$)[a]		
			AUC	Sensitivity (%)	Specificity (%)	Accuracy (%)	Sensitivity (%)	Specificity (%)	Accuracy (%)
Serum	CEA	1.65	0.85	79	88	82	76 (58–89)	60 (32–84)	71 (56–83)
	CA199	12.9	0.74	64	84	70	52 (34–69)	87 (61–98)	63 (48–77)
EV single markers	EGFR	0.23	0.67	59	84	66	52 (34–69)	87 (60–98)	63 (48–77)
	EpCAM	0.08	0.91	86	100	90	85 (68–95)	93 (68–100)	88 (75–95)
	CD24	0.60	0.77	55	100	69	39 (23–58)	100 (78–100)	58 (43–72)
	GPA33	0.02	0.65	47	96	61	42 (25–61)	73 (45–92)	52 (37–67)
	CD133	0.03	0.70	66	76	69	58 (39–75)	53 (27–79)	56 (41–71)
EV weighted combination	EGFR + EpCAM	2.69	0.94	90	100	93	88 (72–97)	100 (78–100)	92 (80–98)
	EGFR + CD24	3.15	0.86	60	90	71	45 (28–64)	100 (78–100)	63 (47–76)
	EpCAM + CD24	4.23	0.97	91	100	94	88 (72–97)	100 (78–100)	92 (80–98)
	EGFR + EpCAM + CD24	4.23	0.97	91	100	94	88 (72–97)	100 (78–100)	92 (80–98)
	EGFR + EpCAM + CD24 + CD133	6.15	0.99	95	100	96	88 (72–97)	93 (68–100)	90 (77–97)
	EGFR + EpCAM + CD24 + GPA33	4.96	0.98	97	100	98	94 (80–99)	100 (78–100)	96 (85–99)
	EGFR + EpCAM + CD24 + GPA33 + CD133	5.63	0.99	98	100	99	100 (89–100)	93 (68–100)	96 (89–100)

[a]Ninety-five per cent confidence intervals are indicated in parentheses.

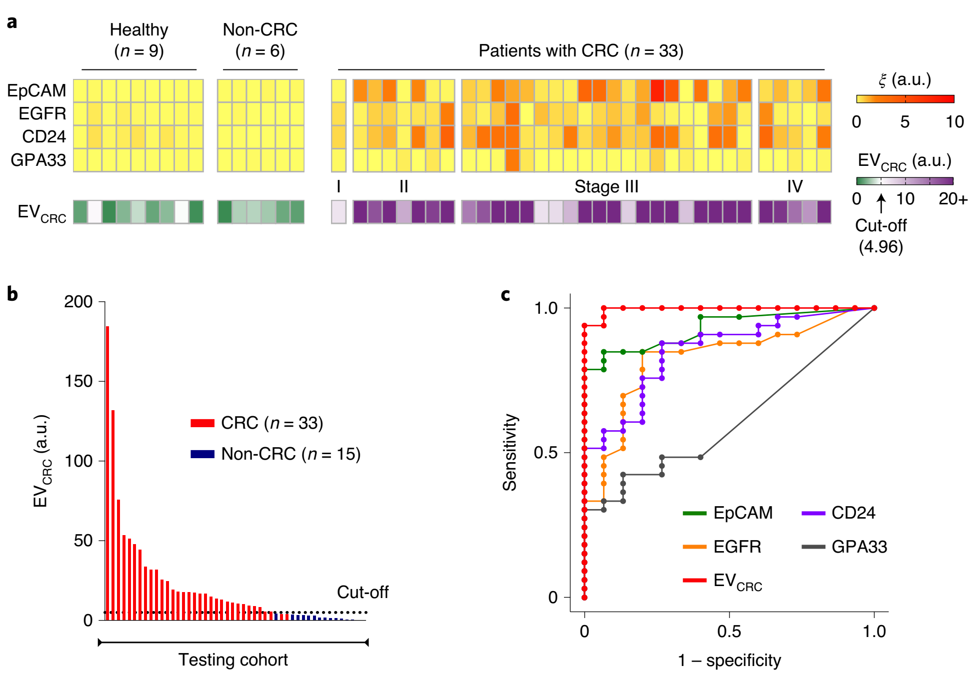

Fig. 4 | Analyses of prospective cohorts for CRC diagnosis. a, Top: plasma EVs from 33 patients with CRC, 9 healthy donors, and 6 patients with non-CRC diseases (gastrointestinal stromal tumour, $n=2$; small bowel internal herniation, $n=2$; appendicitis, $n=2$) were analysed for the expression of EpCAM, EGFR, CD24 and GPA33. Bottom: EV_{CRC} was calculated according to the same formula as the training cohorts. **b**, EV_{CRC} levels were significantly higher in patients with CRC than in the non-CRC controls, which validates the EV-based diagnostic algorithm. The same cut-off value (4.96) from the training set was applied. **c**, EV_{CRC} remained superior in CRC detection, with its AUC significantly larger than those of single markers. Detailed statistics are in Table 2.

samples of recurrent ($n=3$) and non-recurrent ($n=3$) patients were processed. Captured EVs were then lysed and their RNA contents were sequenced for quantitative gene expression profiling (Methods). Principal component analysis, on the basis of mRNA reads, separated the two cohorts (that is, recurrent versus non-recurrent), and more genes were upregulated in the recurrent patient group

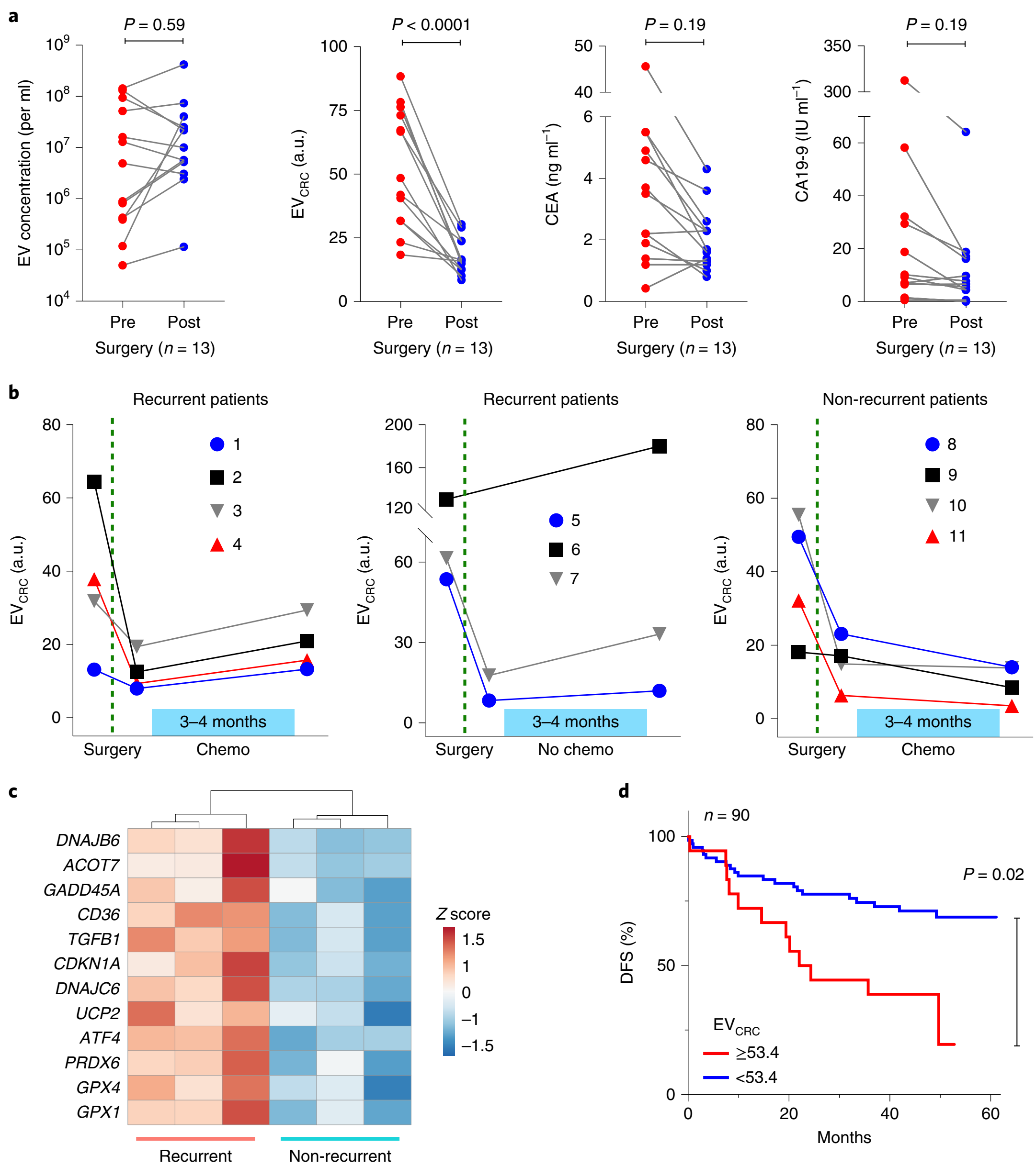

Fig. 5 | HiMEX analyses of longitudinal samples from patients with CRC. a, Blood samples from patients with CRC were analysed before and after surgery ($n=13$). Molecular EV profiling revealed that EV_{CRC} values decreased in all patients ($P<0.0001$, paired two-sided t-test) (middle left), whereas total EV concentrations ($P=0.59$, paired two-sided t-test) (left) and the levels of the conventional serum markers CEA ($P=0.19$, paired two-sided t-test) (middle right) and CA19-9 ($P=0.19$, paired two-sided t-test) (right) showed no significant changes. Each data point in the graphs represents the mean value from technical duplicates. **b**, Plasma EVs were further monitored after surgery, as some patients underwent chemotherapy. EV_{CRC} values rebounded in all patients ($n=7$) with recurrent tumours (left and middle). By contrast, in patients without recurrent tumours ($n=4$) (right), EV_{CRC} values continued to decrease or stabilized from the post-surgery level. Data are mean ± s.e.m. from technical duplicates. **c**, EVs from patients with recurrent ($n=3$) and non-recurrent ($n=3$) CRC were analysed for mRNA expression. Highly variable genes were plotted. Genes that are involved in DNA repair, protection against oxidative pressure and chemoresistance (5-FU) showed higher levels in EVs from patients with recurrent tumours. **d**, Patients with CRC ($n=90$) were monitored for up to five years for DFS. The Kaplan–Meier estimator for DFS was plotted, stratified by preoperative EV_{CRC} levels. High EV_{CRC} values (≥53.4) were associated with poor prognostics ($P=0.02$, two-sided log-rank test).

(Supplementary Fig. 19a). We conducted differential analysis to identify highly variable genes across samples (Supplementary Fig. 19b). Notably, selected genes had higher expression in the recurrent patients (Fig. 5c). These genes are involved in DNA damage repair (*GADD45A* and *CDKN1A*)[42,43], protection against oxidative stress (*GPX1*, *GPX4*, *PRDX6* and *UCP2*)[44–46], fatty acid oxidation (*ACOT7* and *CD36*)[44,47] and resistance to 5-FU (*TGFB1*, *DNAJB6*, *DNAJC6* and *ATF4*)[48–51].

Assessing CRC-EV signature for prognosis of five-year DFS. Finally, we evaluated the ability of EV analyses to predict DFS. High CRC-EV burden may indicate large primary lesions or malignant phenotypes; it is thus conceivable that high levels of EV_{CRC} before surgery could correlate with the risk of disease recurrence, probably caused by either residual tumour presence after surgery or metastasis[52]. We prospectively followed 90 patients with CRC (Supplementary Table 3), who all had preoperative blood tests and underwent tumour resection, for up to five years. From EV analyses, we found that the initial EV_{CRC} scores were higher in the patient group whose tumours eventually progressed into metastasis ($P=0.041$, unpaired two-sided t-test) (Supplementary Fig. 20). Survival analyses further confirmed the association between high EV_{CRC} values and poor DFS. We stratified patients into two groups according to their EV_{CRC} scores, with a score threshold of 53.4 selected to maximize the absolute log-rank statistic between the two strata[53]. We found that 13 out of the 18 patients with high EV_{CRC} scores reported tumour metastasis over the five years of follow-up, compared with 21 out of the 72 patients with EV_{CRC} scores below the threshold; comparing the two survival curves suggested significant prognostic differences between the two groups ($P=0.02$, two-sided log-rank test) (Fig. 5d and Methods). By contrast, preoperative CEA concentrations (cut-off value of 7 ng ml^{-1}), which are often used for CRC prognostics[22,54], displayed borderline performance with respect to DFS in this cohort ($P=0.07$, two-sided log-rank test) (Supplementary Fig. 21).

Discussion

Analysing tumour-derived EVs can provide real-time snapshots of the molecular makeup of tumours, potentially offering timely, better-informed opportunities for clinical intervention. We developed HiMEX to position such EV-based testings closer to clinical reality. The HiMEX approach integrates EV isolation and protein detection in a continuous workflow to enable batch-processing of clinical samples. In particular, HiMEX offers the following technical advantages: (1) direct use of plasma samples for target specific EV protein analyses; (2) superior detection sensitivity (about 1,000-fold higher than that of ELISA), by magnetic enrichment and enzymatic signal amplifications; and (3) a compact device that executes parallel measurements without using mechanical scanners. To prove the concept, we implemented a compact HiMEX device that uses a 96-electrode array. We also showed that the entire HiMEX assay, from EV isolation to analysis, was complete within 1 h.

HiMEX can make EV profiling simple in standard laboratory settings; the technology is cost-effective and capable of fast, parallel EV protein detection. Immunomagnetic pulling—the initial assay step in HiMEX—can also facilitate EV collection for other molecular assays. For example, in surveying drug-resistance status, we immunomagnetically capture EVs directly from plasma; a portion of bead-captured EVs were then used for CRC protein detection by HiMEX, and the rest for targeted mRNA sequencing. To improve the system further, detection electrodes could be engineered to enhance assay sensitivity and throughput. Miniaturizing electrodes into the micrometre scale will improve the mass-detection limit (by reducing sample volumes) and facilitate the construction of dense arrays. Detection electrode surfaces can also be textured with nanostructures to achieve exquisite sensitivity[55]. These improvements may empower HiMEX to detect even scarce EV targets (for example, mutated proteins inside vesicles, protein modification and nucleic acids), thereby enabling comprehensive molecular profiling in a single assay system.

To use HiMEX in CRC detection, we initially performed a bioinformatic survey to determine overexpressed proteins in CRC tissue. Although many proteins have been associated with CRC, we narrowed them down to a panel consisting of EGFR, EpCAM, CD24 and GPA33. Notably, the CRC-EV panel achieved high accuracy in our discovery (98%) and testing (96%) cohorts, which could be attributed to avid release of tumour EVs into the circulation. Moreover, serial EV profiling revealed a correlation between the molecular signature of EV and temporal changes in tumour burden. CRC-EV levels dropped in all patients immediately after surgery; for those patients without near-term recurrence, CRC-EV levels continued to decrease over time, whereas for those patients with recurrence, CRC-EV levels rebounded with tumour progression. We further observed that the initial EV burden can be a predictive risk factor of short-term (five-year) tumour relapse. Combined, these outcomes would widen the potential of EVs as CRC biomarkers, not only for diagnostics but also for treatment monitoring and prognosis.

Several limitations of the current work, however, need to be addressed in future studies. Our comparison between tissue and circulating EVs was carried out for three markers (EpCAM, EGFR and CD24), and was restricted by insufficient amounts of tissue for immunohistochemistry optimization. Expanding both markers (for example, CD133 and GPA33) and sample numbers will firmly establish the presence of CRC-derived EVs as representative of tumour cells. Serial EV tracking, particularly for treatment monitoring, also requires more frequent sampling and larger prospective cohorts than the current study, to obtain robust statistics. As for the patient class, we enrolled resectable patients with established disease, because we were interested in monitoring EV changes during treatment. Invariably, many of our patients were at stages II and III, with only a few at stage I. In future research, it may be interesting to adapt a panel for a CRC-screening tool. Additional molecular markers (for example, KRAS(G12D), KRAS(G13D), KRAS(G12V), BRAF(V600E), MLH1, MSH3, MSH6 and IGF2)[56–58] could be tested for this purpose.

Going forwards, we envision expanding the current study to broaden its impact. First, we could enlarge the study cohort, both patients with CRC and controls, to validate EV-based CRC diagnostics further. Acquiring samples from patients with CRC from multi-institutions would enable us to account for ethnic and geographical diversities; including non-CRC patients who have potentially confounding bowel symptoms (for example, irritable bowel syndrome or Crohn's disease) would serve to refine EV biomarkers and their cut-offs. With its throughput and low equipment cost, HiMEX will expedite processing such sample volumes even in standard laboratory settings. Second, we could design EV tests to discriminate among different CRC stages to complement conventional imaging. Improved non-invasive CRC staging and molecular profiling would be a powerful tool to (1) select patients for neoadjuvant rather than adjuvant chemotherapies; (2) identify groups at high risk for relapse; and (3) select patients for targeted clinical trials. Achieving such classification power will require larger numbers of patients at each CRC stage than in the current study. Third, we could expand EV screening to obtain diverse clinically relevant information. For example, tumour-derived EVs have been shown to present distinct patterns of integrins; these EVs can be taken up by resident cells in an organ-specific manner, preparing pre-metastatic niches[59,60]. Others have posited that chemotherapy can further trigger a primary tumour to release EVs with pro-metastatic potential[61,62]. The ability of HiMEX to enrich tumour-specific EVs will facilitate the detection of these EV subpopulations to help understand, monitor and even predict metastasis.

Methods

Collecting clinical samples. The study was approved by the Institutional Review Board (IRB) of Kyungpook National University Medical Centre (principal investigator: J.S.P.) where all clinical samples were collected. The procedures followed were in accordance with institutional guidelines. Informed consent was obtained from all subjects. Peripheral blood (~15 ml) was withdrawn from patients and centrifuged at 400g for 15 min to separate plasma from red blood cells and buffy coat. We used 20 µl of plasma to analyse each marker.

Constructing the HiMEX system. The HiMEX device consisted of a microcontroller (ATSAMD21G18, Atmel Corporation), a digital-to-analogue converter (DAC8552, Texas Instruments), an analogue-to-digital converter (AD7490, Analog Devices), a multiplexer (ADG708, Analog Devices), and 96 potentiostats. Each potentiostat had two operational amplifiers (AD8606, Analog Devices): one amplifier maintained the potential difference between a working and a reference electrode, and the other one functioned as a transimpedance amplifier to convert current to a voltage signal. The current-measuring range of the transimpedance amplifier was ±7.5 µA. A 12 × 8 electrode array (96×220, DropSens) was used.

Preparing assay reagents. The first step was to produce the immunomagnetic beads. 5 mg of magnetic beads with epoxy groups (for example, 14302D, Dynabeads M-270 Epoxy, Invitrogen) were suspended in 1 ml of 0.1 M sodium phosphate solution at room temperature for 10 min. Magnetic beads were separated from the solution with a permanent magnet and re-suspended in 100 µl of the same solution. Then, 100 µg of antibodies against CD63, CD9 or CD81 (see Supplementary Table 1 for details) was added and mixed thoroughly. Next, 100 µl ammonium sulfate solution (3 M) was added, and the whole mixture was incubated for 2 h at room temperature and then overnight at 4 °C with slow tilt rotation. Beads were washed twice with PBS and finally re-suspended in 2 ml of PBS with 1% bovine serum albumin (BSA). The second step was to label antibodies. 10 mM sulfo-NHS-biotin (A39257, Pierce) solution in PBS was incubated with antibodies for 2 h at room temperature. Unreacted sulfo-NHS-biotin was removed using a Zeba spin desalting column, 7 K MWCO (89882, Thermo Scientific). Biotinylated antibodies were kept at 4 °C until use.

HiMEX assay. We mixed EV samples (10 µl of EV-spiked PBS or 20 µl of plasma) with 50 µl of the immunomagnetic bead solution for 15 min. We then washed the beads by collecting them with a permanent magnet, discarded the supernatant and added a fresh buffer (50 µl, PBS with 1% BSA). Then, 10 µl of biotinylated antibodies of interest (20 µg ml^{-1} in PBS) was added and the mixture was incubated for 15 min. We separated and washed the beads as described above and added 5 µl of streptavidin-conjugated HRP enzymes (21130, Pierce, 1:100 diluted in PBS; 15 min at 21 °C). Beads were separated, washed and finally suspended in 7 µl of PBS. To generate the signal, the prepared bead solution and 20 µl of UltraTMB solution (34028, ThermoFisher Scientific) were loaded on top of the screen-printed electrode. After 3 min, the chronoamperometry measurement was started. The current levels between 50 and 55 s were averaged. The total assay time was about 1 h. All assay steps were carried out at 21 °C.

EV capture. Antibodies for EV capture (anti-CD63, anti-CD9 or anti-CD81) were coupled to magnetic beads, at a ratio of 10 µg of total antibody to 100 µl of beads, by overnight incubation at 4 °C with rotation. Beads were washed three times with 500 µl of PBST buffer (PBS plus 0.001% Tween 20) and resuspended in 100 µl of the same buffer. EVs were isolated by OptiPrep density gradient ultracentrifugation as previously described[63]. The final pellet was resuspended in 100 µl of PBS. Collected EVs (5 µl, 5×10^9 vesicles ml^{-1}) were then mixed with 25 µl of antibody-coated beads, followed by addition of 170 µl of PBST and incubation at 4 °C with rotation. Beads with pulled-down EVs were collected and washed three times with 500 µl PBST. The flow-through was concentrated to 50 µl using Amicon Ultra 10-kDa filters (UFC501096, Millipore Sigma). For western blotting, samples were lysed with non-reducing LDS sample buffer, boiled for 5 min at 95 °C and loaded on a gel. Proteins were separated by SDS–PAGE, transferred to a nitrocellulose membrane and immunostained with the following antibodies: anti-CD63 (clone H5C6, BD Biosciences, 1:200 dilution); anti-CD9 (clone D8O1A, Cell Signaling, 1:1,000 dilution); and anti-CD81 (clone 1.3.3.22, Thermo Fisher, 1:500 dilution). HRP-conjugated secondary antibodies were added, then chemiluminescence substrate (WesternBright Sirius, Advansta), and blots were developed using autoradiographic films (Fig. 2b and Supplementary Fig. 22). Films were digitized and quantification of signal intensity was performed using ImageJ version 1.51s.

ELISA. CD63 antibody (Ancell) and IgG1 antibody (Ancell) were diluted in PBS (5 µg ml^{-1}) and transferred to the Maxisorp 96-well plate (Nunc) for overnight incubation at 4 °C. After washing with PBS, 2% BSA in PBS blocking solution was added to the plate (1 h incubation at room temperature). Subsequently, EV samples (in 100 µl PBS) were added to each well for a 1-h incubation at room temperature. After discarding the blocking solution, antibodies (1 µg ml^{-1}) against various markers were inserted in each well and incubated for another 1 h at room temperature. Unbounded antibodies were triple-washed with PBS. Streptavidin-HRP molecules then were added to each well, and the mixture was incubated for 1 h at room temperature. After washout with PBS, the chemiluminescence signal was measured by a plate reader (Tecan).

Flow cytometry. About 10^6 cells per marker were used for flow cytometry experiments. Cells were fixed with 4% paraformaldehyde for 10 min at room temperature and then washed with PBS (0.5% BSA). Next, cells were blocked with BSA (0.5% in PBS) and incubated with primary antibodies (4 µg ml^{-1}). Labelled cells were washed, incubated with fluorophore-conjugated secondary antibodies (2 µg ml^{-1}; Abcam), and washed again. Control samples were similarly labelled using isotype-matched IgG and secondary antibodies. As a background estimator, an aliquot of cells was incubated with secondary antibodies only. Blocking and incubation with antibodies (primary and secondary) were performed for 30 min each at room temperature. Every washing step consisted of three 5-min washes at 300*g* with PBS (0.5% BSA). Fluorescence signals from the labelled cells were measured using BD LSRII Flow Cytometer (BD Biosciences). Mean fluorescent intensities (MFIs) were obtained from three types of sample (that is, targeted, IgG and secondary-only). The expression level of a target marker was obtained as $(MFI_{target} - MFI_{IgG})/MFI_{secondary}$. Gating information is shown in Supplementary Fig. 23.

Cell culture. A panel of CRC cell lines was purchased (ATCC) and grown in the vendor-recommended medium: DLD-1 (RPMI-1640, Cellgro); HT29 (MacCoy's 5a modified with 2% $NaHCO_3$, Cellgro); HCT116 (MacCoy's 5a, Cellgro); Colo201 (RPMI-1640 modified with 1% sodium pyruvate, Cellgro); SW480 and SW620 (DMEM, Cellgro); CCD-18Co (Eagle's MEM, Cellgro); CCD-112CoN (Eagle's MEM, Cellgro); CCD-33Co (Eagle's MEM, Cellgro). All medium was supplemented with 10% FBS and penicillin–-streptomycin (Cellgro). All cell lines were tested and determined to be free of mycoplasma contamination (MycoAlert Mycoplasma Detection Kit, Lonza, LT07-418).

Isolating EVs from cell culture. For spike-in experiments, we prepared pure EVs from cell culture. Cells at passages 1–15 were cultured in vesicle-depleted medium (with 5% depleted FBS) for 48 h. Conditioned medium from approximately 10^7 cells was collected and centrifuged at 300*g* (5 min). Supernatant was filtered through a 0.2-µm membrane filter (Millipore) and concentrated by centrifugation at 100,000*g* (1 h). After the supernatant was removed, the EV pellet was washed with PBS and centrifuged at 100,000*g* (EV). Collected EVs were resuspended in PBS and stored at 4 °C. For the stock solution samples, EV concentrations were estimated through nanoparticle tracking analysis (NTA).

Immunohistochemistry. Tumoural and non-neoplastic tissue sections from patients with CRC were subjected to immunohistochemical staining (Fig. 3a and Supplementary Fig. 24). Primary monoclonal antibodies against EGFR (EGFR1, Abcam, 1:50 dilution), EpCAM (VU-1D9, Abcam, 1:100 dilution) and CD24 (eBioSN3, ebioscience, 1:100 dilution) were used and the staining was conducted on a Ventana BenchMark XT automated slide stainer (Ventana Medical Systems), according to the manufacturer's recommendations. Stained images were scored, on the basis of the fraction of positive cells among total cancer cells, by a pathologist (G.Y.) who was blinded to clinicopathological variables and EV profiling results. Positive cells were defined by positively stained cytoplasmic or membranous pattern within cancer cells in the face of concurrent negative labelling in non-neoplastic tissues. The marker expression was ranked from 0 (negative) to 1 (positive) with increments of 0.1; this level was used as an ordinal variable in Spearman's rank test with EV profiling results.

Tumour recurrence status determination. Recurrence was diagnosed pathologically by either surgical resection or radiological detection of lesions that increased in size over time. Radiologists and pathologists independently assessed the radiological imaging and pathological specimens. Local recurrence was defined as any recurrence within the pelvic cavity or the perineum. Systemic recurrence was defined as any recurrence outside the pelvic cavity.

EV mRNA analyses. We used about 3 ml of patient plasma samples as recommended by the vendor. EVs from patient plasma were captured on magnetic beads and RNA was isolated using exoRNeasy Serum/Plasma Starter kit (77023, Qiagen). Libraries were prepared with QIAseq Targeted RNA panel according to the manufacturer's instructions (Human Molecular Toxicology Transcriptome, Qiagen). Libraries were quantified using Qubit dsDNA HS Assay kit (Q32850, ThermoFisher Scientific) and QIAseq library Quant Assay kit, and the library size was determined using Agilent 2100 Bioanalyzer (High Sensitivity DNA Analysis Kit, Agilent Technologies). Prepared libraries were pooled (4 nM for each library), and the pooled mixture (6.5 pM) was run on an Illumina MiSeq (MiSeq Reagent Kit v2, Illumina). Data were analysed using the DESeq2 package version 1.30.1 in R version 3.6.1.

Single EV imaging. EVs from CRC cell lines (SW480, HT29; 10^8 vesicles ml^{-1}) were incubated at room temperature for 30 min on a glass slide (63429-04, Electron Microscopy Science). The solution was drained with a paper towel followed by incubation for 15 min in 4% paraformaldehyde solution (AAJ19943K2, Thermo Scientific) at room temperature for EV fixation. The slide was washed twice with PBS. Perm/Wash buffer (554723, BD Science) was added for EV permeabilization, and the slide was incubated for 5 min at room temperature. After five washes in PBS, the slide was blocked with 2% BSA in PBS for 30 min. Then, 10 µl of AF647 (1434, Click Chemistry Tools)-labelled anti-CD63 (215-020, Ancell), CD9 (555675, BD science) or CD81 (555370, BD Science) antibodies (10 µg ml^{-1} in PBS with 1% BSA) was added for EV labelling overnight at 4 °C. After three washes in PBS, the slide was covered with a coverslip and EVs were imaged with BX63 Fluorescent microscope (Olympus).

Bioinformatic algorithm for selecting initial markers. From the Human Protein Atlas portal, we downloaded immunohistochemistry data on CRC and normal tissues. Numerical values were assigned to staining levels (high, 3; medium, 2; low, 1; undetected, 0) and a mean staining value was obtained for each protein. For each tissue type (that is, CRC and normal), these values were normalized to *z*-scores. Markers were then ranked according to their differential *z*-score (CRC – normal). This list was narrowed down by cross-referencing UniProt to select proteins at specific cellular domains (for example, transmembrane). We further filtered the list using EV databases to collect markers reportedly found in EVs. Analyses were performed in R using custom-written scripts.

Statistical analyses. Statistical analysis was performed using Stata version 12.1 (StataCorp LP), GraphPad Prism version 8.0 (GraphPad Software Inc.), or R version 3.6.1. For all statistical tests, P values <0.05 were considered significant.

Marker selection for CRC detection. We selected candidate predictive markers from the EV profiling data (Supplementary Fig. 12a) by applying the least absolute shrinkage and selection operator (Lasso) regression. Note that we augmented non-CRC controls with EV data from healthy plasma samples. We calculated the cross-validation error and determined the tuning parameter (λ) that minimized cross-validation error. We used the glmnet package version 4.1 in R.

CRC diagnosis. To assess the ability of the selected protein markers, either individually or in combination, to predict CRC cases, we conducted an ROC analysis using the training cohort ($n=83$). The specific individual biomarkers under consideration were EGFR, EpCAM, CD24, GPA33 and CD133. We also considered the the linear combination of two markers (EGFR + CD24, EGFR + EpCAM, and EpCAM + CD24), three markers (EGFR + EpCAM + CD24), four-marker combinations (EGFR + EpCAM + CD24 + GPA33 and EGFR + EpCAM + CD24 + CD133), and five-marker combinations (EGFR + EpCAM + CD24 + GPA33 + CD133). For each biomarker combination, optimal weights were determined via logistic regression. We constructed ROC curves and determined level cut-off values that maximized the sum of sensitivity and specificity. Standard formulas were used to define sensitivity (true positive rate), specificity (true negative rate), and accuracy [(true positive + true negative)/(positive + negative)]. AUCs were compared following Delong's method[64]. We next analysed the prospective cohort ($n=48$) using the selected cut-off values, and determined the classification performance of each biomarker and biomarker combination. Exact 95% confidence intervals for the sensitivity, specificity and accuracy were also calculated. Analyses were performed using the pROC package version 1.17.0.1 in R (ref. [65]).

Survival analyses. We monitored 90 patients with CRC, who underwent curative surgery, for up to 61 months. DFS was defined as the interval between surgery and either the first radiological recurrence or death as a result of CRC. We used the Kaplan–Meier method to calculate the survival functions stratified by EV_{CRC} or CEA levels. The chosen EV_{CRC} cut-off value for patient classification was the value that maximized the absolute log-rank score statistic between the two classification groups: the P value for comparing the corresponding survival curves for the two groups above versus below the selected EV_{CRC} cut-off value was obtained using conditional Monte Carlo simulation with 10,000 replications to approximate the null distribution of the maximally selected rank statistic[53]. The log-rank test was used to compare survival curves under the conventional CEA cutpoint. Analyses were performed using the maxstat package version 0.7.25 (ref. [66]) and the survival package version 3.2.11 in R.

Reporting Summary. Further information on research design is available in the Nature Research Reporting Summary linked to this article.

Data availability
The main data supporting the results in this study are available within the paper and its Supplementary Information. The raw patient datasets generated and analysed during the study are available from the corresponding authors on reasonable request, subject to approval from the Institutional Review Board of the Kyungpook National University Hospital. For initial marker selection, we used the public databases, the Human Protein Atlas (https://www.proteinatlas.org) and UniProt (https://www.uniprot.org). Source data are provided with this paper.

Code availability
Source codes for the marker selection are available at https://csb.mgh.harvard.edu/bme_software.

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Author contributions

J.P., J.S.P., C.M.C., R. Weissleder and H. Lee designed the study, prepared the figures and wrote the manuscript. J.S.P. and G.-S.C. collected patient samples. G.Y. performed tissue immunohistochemistry staining. C.-H.H. developed the HiMEX system. J.P., A.J., H.-Y.L., J.V.D., H. Li and J.M. performed research and analysed data. L.W., L.B. and B.S.C. guided mRNA analysis. J.S.P., G.-S.C., C.M.C. and R. Weissleder analysed clinical data. K.C., R. Wang and H. Lee performed statistical analyses.

Competing interests

The authors declare the filing of a patent (US20190346434A1) that was assigned to and handled by Massachusetts General Hospital. The following disclosures are not related to the subject matter of this work. R.Weissleder is a consultant to ModeRNA, Tarveda Pharmaceuticals, Lumicell, Seer, Earli, Alivio Therapeutics, Aikili Biosystems and Accure Health. H. Lee is a consultant to Exosome Diagnostics, Accure Health and Aikili Biosystems.

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☒	☐	For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
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Software and code

Policy information about availability of computer code

Data collection	Flow Jo (flow cytometry)
Data analysis	Graphpad Prism (ver 8); Stata (ver. 12.1); R (ver 3.6.1); ImageJ (ver. 1.51s); DESeq2 (ver. 1.30.1); maxstat (ver. 0.7-25)

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The main data supporting the results in this study are available within the paper and its Supplementary Information. The raw patient datasets generated and analysed during the study are available from the corresponding authors on reasonable request, subject to approval from the Institutional Review Board of the Kyungpook National University Hospital. For initial marker selection, we used public databases: Human Protein Atlas (https://www.proteinatlas.org) and UniProt (https://www.uniprot.org).

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Life sciences study design

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Sample size	Discovery cohort: Sample-size was not predetermined, but the sample numbers are powered to determine whether the EV-based CRC test offers high diagnostic accuracy. We used the area under the ROC curve (AUC) as an index of accuracy. Our null hypothesis was that the EV-CRC test has an AUC of 0.7 (a fair test), whereas the alternative hypothesis was that the EV-CRC test performs significantly better (AUC = 0.9, a highly accurate test). The sample sizes we used (58 CRC patients and 25 controls) achieves 95% power to detect this AUC difference of 0.2 at a significance level of 5% (two-sided z-test). Validation cohort: The sample size (33 CRC and 15 controls) was powered to estimate the lower limit of detection, sensitivity and specificity. Assuming an assay sensitivity of 0.95, the lower confidence limit is 0.80 (at a significance level of 5% and with 80% power). Similarly, assuming an assay specificity of 0.95, the lower confidence limit is 0.71 (at a significance level of 5% and with 80% power).
Data exclusions	No data were excluded.
Replication	All clinical samples were measured at least twice (technical replica) for each protein marker, and the mean values were used for analyses. All attempts at replication were successful. For cancer diagnostics, we first defined a statistical model using a training set, and then validated the model using a testing set.
Randomization	Not relevant. Patients who met the eligible criteria (patients had to have a diagnosis of colorectal cancer from tissue biopsy) and gave informed consent were recruited on a rolling basis. To control for covariates, ratio of gender, ratio of normal control to CRC, and stage of cancer were considered.
Blinding	The study was double-blinded during data collection. Sample information (that is, cancer vs. control, recurrent vs. non-recurrent) was not known to the investigators performing EV analyses, cellular protein analysis and IHC staining. The EV test results were not known to the clinical investigators.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study	
☐	☒	Antibodies
☐	☒	Eukaryotic cell lines
☒	☐	Palaeontology
☒	☐	Animals and other organisms
☐	☒	Human research participants
☒	☐	Clinical data

Methods

n/a	Involved in the study	
☒	☐	ChIP-seq
☐	☒	Flow cytometry
☒	☐	MRI-based neuroimaging

Antibodies

Antibodies used	We provide comprehensive information on antibodies in Supplementary Tables 1 and 4.
Validation	All antibodies used in this study were validated from commercial vendors. Validation statements are on the manufacturer's website.

Eukaryotic cell lines

Policy information about cell lines

Cell line source(s)	Colorectal cancer cell lines DLD-1(CCL-221), HT29(HTB-38), HCT116(CCL-247), Colo201(CCL-224), SW480(CCL-228) and

Cell line source(s)	SW620(CCL-227), and colorectal normal cell lines CCD-18Co(CRL-1459), CCD-112CoN(CRL-1541) and CCD-33Co(CRL-1539), were purchased from American Type Culture Collection (ATCC) within the last three years.
Authentication	All cell lines had been authenticated by ATCC, and were used without any modification.
Mycoplasma contamination	All cell lines were tested and determined to be free of mycoplasma contamination (MycoAlert Mycoplasma Detection Kit, Lonza, LT07-418).
Commonly misidentified lines (See ICLAC register)	No commonly misidentified cell lines were used in the study.

Human research participants

Policy information about studies involving human research participants

Population characteristics	Subjects considered for enrollment in this study were evaluated solely on the basis of protocol eligibility criteria (patients had to have a diagnosis of colorectal cancer from tissue biopsy). Summaries of patient characteristics are provided in Table 1 and Supplementary Table 2. Among the 142 clinical samples, the characteristics of the participants were as follows: Age (years): Median 61, 18 ≤ age ≤ 82 for CRC diagnostics; median 65, 34 ≤ age ≤79 for treatment monitoring. Gender: male (78, 60%); female (53 ,40%) for CRC diagnostics; male (9, 82%); female (2 ,18%) for treatment monitoring.
Recruitment	The first subjects who met eligibility criteria and gave informed consent were enrolled. Children defined as age less than 18 were excluded from the study. Otherwise, there were no specific exclusion criteria. Considering that CRC samples were collected in Korea, there could be a potential bias of representing an ethnically Asian population.
Ethics oversight	The study was approved by the Institutional Review Board (IRB) of Kyungpook National University Medical Center and by the IRB of Massachusetts General Hospital.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Flow Cytometry

Plots

Confirm that:

- [x] The axis labels state the marker and fluorochrome used (e.g. CD4-FITC).
- [x] The axis scales are clearly visible. Include numbers along axes only for bottom left plot of group (a 'group' is an analysis of identical markers).
- [ ] All plots are contour plots with outliers or pseudocolor plots.
- [ ] A numerical value for number of cells or percentage (with statistics) is provided.

Methodology

Sample preparation	All cell lines were purchased from ATCC and grown in the vendor-recommended media. About 1 million cells per marker were used for flow cytometry. Cells were fixed with 4% paraformaldehyde for 10 min at room temperature (RT) and then washed with PBS (0.5% BSA). Next, cells were blocked with BSA (0.5% in PBS) and incubated with primary antibodies (4 μg/mL). Labelled cells were washed, incubated with fluorophore-conjugated secondary antibodies (2 μg/mL; Abcam), and washed again. Control samples were similarly labelled using isotype-matched IgG and secondary antibodies. As a background estimator, an aliquot of cells was incubated with secondary antibodies only.
Instrument	Fluorescence signals from the labelled cells were measured using a BD LSRII Flow Cytometer (BD Biosciences).
Software	Flow Jo was used to analyse the data.
Cell population abundance	10,000 cells per each measurement.
Gating strategy	Single-cell populations (~85%) were gated via the FSC/SSC polygon-gating method. Mean fluorescent intensities (MFIs) were obtained from three types of sample (targeted, IgG, and secondary-only). The expression level of a target marker was obtained as (MFI_target – MFI_IgG)/MFI_secondary.

- [x] Tick this box to confirm that a figure exemplifying the gating strategy is provided in the Supplementary Information.